

The Iterated Prisoner's Dilemma

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1 Introduction

This project will study the Prisoner's Dilemma, with a particular focus on the iterated Prisoner's Dilemma and the behaviour of different strategies when they interact repeatedly.

The current working question is:

How do different strategies affect the outcomes when they are made to repeatedly play the Prisoner's Dilemma under different conditions?

At this stage, the question is intentionally broad. The aim of the initial background research is to understand the different ways in which the Prisoner's Dilemma has been studied and to identify an interesting aspect of the game that can be investigated further.

Possible conditions to consider include the number of rounds, the number of players, the payoff structure, and the set of strategies being used.

2 Background

2.1 The Prisoner's Dilemma

The Prisoner's Dilemma is a two-player game in which each player chooses between cooperation and defection. The payoff received by each player depends on the choices made by both players.

A general payoff matrix can be written as

	C	D
C	(R, R)	(S, T)
D	(T, S)	(P, P)

where the standard Prisoner's Dilemma conditions are

$$T > R > P > S.$$

For example, the payoffs can be represented by

	C	D
C	$(3, 3)$	$(0, 5)$
D	$(5, 0)$	$(1, 1)$

Defection is a dominant strategy in the one-shot Prisoner's Dilemma, since a player receives a higher payoff from defecting regardless of the action chosen by the other player. Consequently, mutual defection is the Nash equilibrium, even though mutual cooperation gives both players a higher payoff.

This conflict between individual incentives and the joint outcome is one of the main reasons the Prisoner's Dilemma is studied.

2.2 The Iterated Prisoner's Dilemma

In the iterated Prisoner's Dilemma, the game is played repeatedly by the same players. This introduces the possibility that decisions in one round can depend on what happened in previous rounds.

A strategy in the iterated Prisoner's Dilemma can therefore be viewed as a rule that determines which action a player takes based on the available history of the game.

For example, a strategy might always cooperate, always defect, or change its action based on the opponent's previous action.

The repeated nature of the game creates behaviour that does not occur in the one-shot version. In particular, players can respond to previous actions, potentially rewarding cooperation or responding to defection.

3 Background Research

3.1 Strategy Performance

One major area of research in the iterated Prisoner's Dilemma concerns the performance of different strategies.

Some simple strategies include:

- **Always Cooperate:** cooperate in every round.
- **Always Defect:** defect in every round.
- **Tit-for-Tat:** cooperate in the first round and then copy the opponent's previous action.

Comparing these strategies raises questions about whether one strategy can consistently outperform others, and how the performance of a strategy depends on the strategies it interacts with.

3.2 Axelrod's Tournaments

Robert Axelrod investigated the iterated Prisoner's Dilemma by organizing computer tournaments in which different strategies played against one another.

One of the important results of these tournaments was the strong performance of Tit-for-Tat. The strategy begins by cooperating, responds to defection by defecting, and returns to cooperation when the opponent does so.

Axelrod's work suggests that a strategy's performance cannot necessarily be understood by considering the strategy in isolation. Instead, its performance depends on the strategies with which it interacts.

This leads to the broader question of whether the success of a strategy is a general property of that strategy or a consequence of the particular environment in which it is being tested.

3.3 Cooperation

Another major area of research concerns the emergence and persistence of cooperation.

The one-shot Prisoner's Dilemma predicts mutual defection under the usual assumptions, yet repeated interactions can produce sustained cooperation between players.

Research into cooperation has considered mechanisms such as reciprocity, where cooperation can be maintained because players expect future interactions with one another.

This raises questions about the conditions under which cooperation can emerge and whether those conditions depend on the strategies being used.

3.4 Exploitation and Strategy Interactions

Different strategies can respond to one another in very different ways. A strategy that performs well against one opponent may perform poorly against another.

This means that studying only the average performance of a strategy may not give a complete picture of its behaviour.

An interesting aspect of the iterated Prisoner's Dilemma is therefore the possibility of one strategy exploiting another. This also raises the question of whether there can be a strategy that performs well against a wide range of opponents, or whether performance is fundamentally dependent on the opponent.

3.5 Population Dynamics

The Prisoner's Dilemma can also be studied using populations of players rather than considering only a single pair of players.

In this setting, the frequencies of different strategies can change over time depending on their payoffs. A strategy that performs well may become more common, while a strategy with lower performance may become less common.

This introduces a different type of question from simply determining which strategy wins a single tournament. Instead, the focus can be placed on how the composition of a population changes over time.

Some studies have also considered situations in which cooperation and defection do not simply converge to one fixed state, but can change cyclically over time.

However, throughout the course of this project, we will only be looking at games involving pairs of players.

3.6 Stability of Outcomes

Another possible way of studying the iterated Prisoner's Dilemma is to consider the stability of the outcomes produced by different strategies.

For example, repeated interaction between two players may eventually produce a constant sequence of outcomes such as

$$(C, C), (C, C), (C, C), \dots$$

while other strategy combinations may produce alternating or otherwise changing outcomes.

This raises the question of what conditions cause repeated interactions to settle into a stable pattern and what conditions instead lead to continued changes in behaviour.

At this stage, stability is being considered as a possible direction for the project rather than as the final research question.

4 Potential Research Directions

The background research suggests several possible directions for further investigation.

- How does the performance of a strategy depend on the strategy of its opponent?
- Under what conditions can cooperation persist?
- Can one strategy consistently exploit another strategy?
- How does changing the payoff structure affect strategy performance?
- How does increasing the number of rounds affect the outcomes of different strategy pairings?

These directions are not intended to represent separate final research questions. They are possibilities that emerged from my current research of the Prisoner's Dilemma.